

Control ROBOTIC CAR With Voice

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This project describes the implementation of a voice-controlled robotic car using Arduino. In this project, the user gives specific voice commands to the robot through an Android app installed on the smartphone. At the receiving side, a Bluetooth transceiver module receives the commands and forwards them to the Arduino on the robotic car.

Arduino controls the movements of the robot according to received commands. The robot moves forwards, backwards, left and right, and stops according to the voice commands forward, backward, left, right and stop, respectively. The authors' prototype of the voice-controlled robotic car is shown in Fig. 1.

The system consists of a trans-

mitter (Android smartphone) and a receiver (robot). Block diagrams of the transmitter and receiver sides are shown in Figs 2 and 3, respectively.

Screenshot of the home screen of Voicecontrol app is shown in Fig. 4.

Circuit and working

The circuit diagram of the receiver side of the robot is shown in Fig. 5.

Major components used in this project are described below.

Android smartphone with app. Android speech-recognition app (voice-control.apk) used here was developed using MIT App Inventor. When the app is running in the smartphone, user's voice commands are detected by the microphone present in the phone.

Commands are processed, and speech-to-text conversion is done within the app using Google's speech-recognition technology. Text is then sent to the receiver side (that is, robotic car) via Bluetooth.

Arduino Uno R3. Arduino Uno is an AVR ATmega328P microcontroller (MCU)-based development board with six analogue input pins and 14 digital

I/O pins. The MCU has 32kB ISP flash memory, 2kB RAM and 1kB EEPROM. The board provides the capability of serial communication via UART, SPI and I2C. The MCU can operate at a clock frequency of 16MHz. In this project, digital I/O pins 2, 3, 4 and 5 of Arduino are configured as output pins. Pins 0 and 1 of Arduino are used

for serial communication with HC-05 Bluetooth module.

Text received via Bluetooth is forwarded to Arduino Uno board using UART serial communication protocol. Arduino program voice_ctrl.ino checks the text received and, if it is a matching string, Arduino controls the movements of the robot accordingly. Voice commands used for controlling the robot and their functions are shown in Table I.

HC-05 Bluetooth module. HC-05 module is an easy-to-use Bluetooth Serial Port Protocol (SPP) module, designed for transparent wireless serial connection setup. Serial port Bluetooth module has a fully-qualified Bluetooth V2.0 + EDR (enhanced data rate) 3Mbps modulation with complete 2.4GHz radio transceiver and baseband.

Hardware features. These are:

- Typical -80dBm sensitivity
- Up to +4dBm RF transmit power
- 3.3V to 5V I/O
- Programmable input/output (PIO) control
- UART interface with programmable baud rate

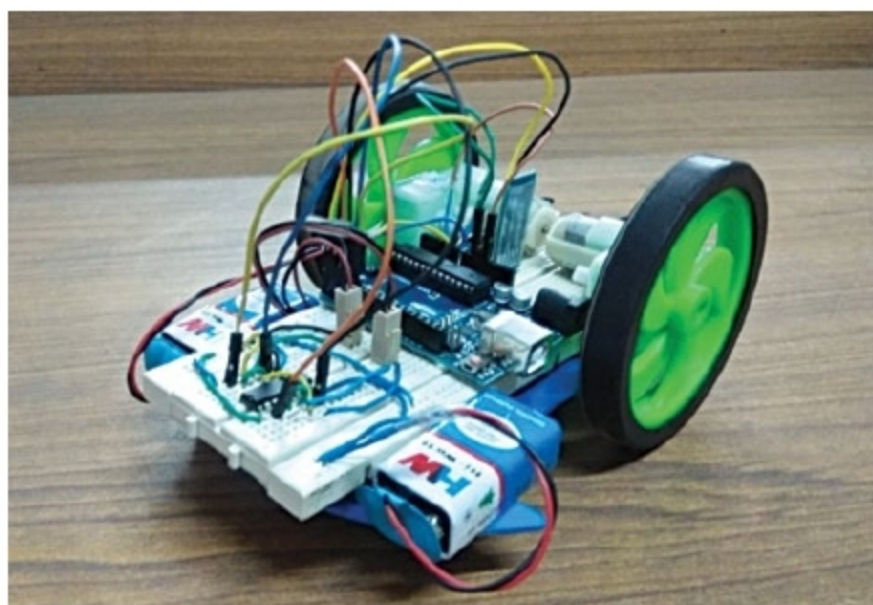


Fig. 1: Authors' prototype of the voice-controlled robotic car

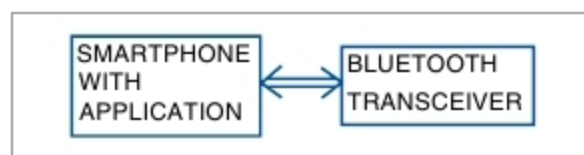


Fig. 2: Block diagram of transmitter side

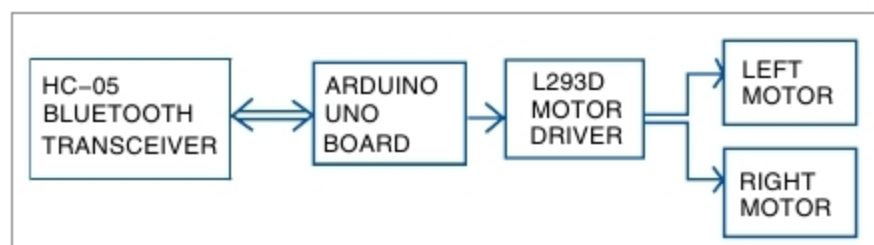


Fig. 3: Block diagram of receiver side

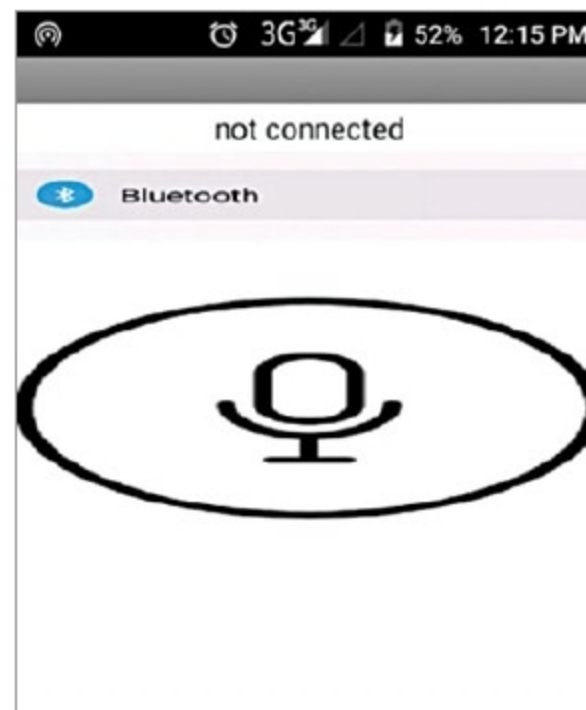


Fig. 4: Screenshot of the home screen of Voicecontrol app